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Exploring celestial classification: Astrophysical features-guided machine learning for spectral and morphological analysis☆

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ABSTRACT

Celestial classification, traditionally based on spectral analysis, helps understand the characteristics and distribution of solar radiation, aiding in the design of solar sail technology and potentially reducing energy costs in space missions. This research investigates the spectral and morphological classification of celestial entities by integrating feature engineering with astrophysical knowledge and principles, utilizing Machine Learning (ML) methodologies. These insights enabled the careful enhancement of the feature set, resulting in the systematic elimination of irrelevant and unstructured data, thereby improving both the model's accuracy and its computing efficiency. The examination of the Sloan Digital Sky Survey (SDSS) dataset highlights redshift and near-infrared measurements (i and z filters) as crucial spectral parameters for classifying stars, galaxies, and quasars. Feature selection streamlined the dataset from 17 initial features to the most pertinent filters (u, g, r, i, z) and redshift, thereby enhancing computational efficiency and model correctness. The Random Forest classifier attained the best accuracy (98%) across all classes by utilizing these features, surpassing both k-nearest neighbors (k-NN) and support vector machines (SVM). For morphological classification, the YOLOv5, YOLOv7, and YOLOv8 models were trained on a tailored dataset to classify galaxies into five morphological categories: Elliptical, Spiral, Irregular, Merging, and Peculiar. Quantitative research indicated that YOLOv8 achieved the highest performance, with 95.5% precision across all galaxy classifications and an overall recall of 73.7%, underscoring its effectiveness in identifying various galaxy morphologies. This comprehensive investigation enhances model interpretability and accuracy, underscores the efficacy of astrophysically motivated features, and establishes a robust framework for real-time large data analysis in astrophysical research, providing a benchmark for industrial applications through advanced data-driven approaches.

1. Introduction

The classification of celestial objects is a cornerstone of astrophysics and is critical for understanding the structure, evolution, and large-scale complexity of the Universe. Astronomical objects encompass any naturally occurring physical entity, group, or structure within the observable Universe, including galaxies, planetary systems, star clusters, nebulae, quasars, exoplanets, galaxy clusters, and pulsars (Narlikar, 1996). Advances in modern observational technology have greatly enhanced astronomers' ability to collect and analyze data (Tyson, 2010), leading to the emergence of the field of astroinformatics (Borne, 2010), which focuses on the efficient analysis of large-scale astronomical datasets, often through the automated classification of astronomical images (Ackermann et al., 2018). Spectral classification is commonly used to categorize celestial objects according to their luminosity, emission and absorption lines, and redshift (Petrillo et al., 2019), providing information on their composition, temperature, and relative velocity. Spectral data are typically represented as numerical measurements of light intensity across different wavelengths. Complementary to this, morphological taxonomy classifies celestial objects based on their visual shape and structural features derived from telescopic imaging (Zhu et al., 2019). The morphological classification of celestial objects, particularly galaxies, remains a central problem in astronomy, as it is fundamental to constraining models of cosmic structure formation and evolution (Zhu et al., 2019).

With the rapid evolution of observational and computational technologies, the need for innovative analytical methodologies that can

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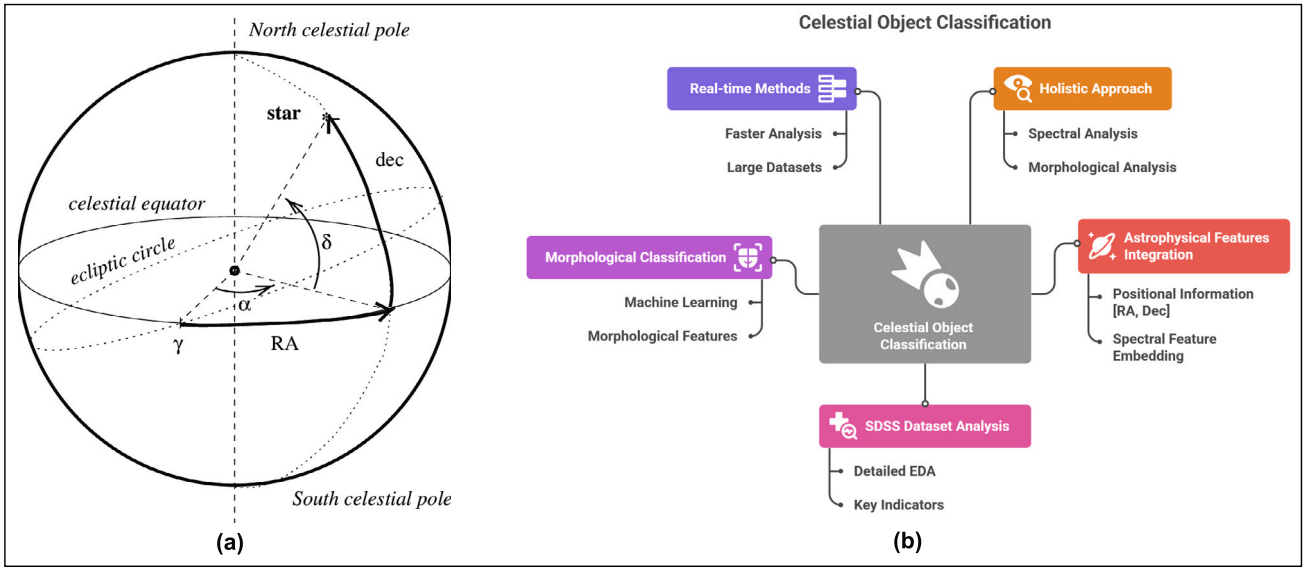


Fig. 1. (a) Celestial coordinate framework defining right ascension (RA) and declination (Dec) used to describe the spatial position of astronomical sources. (b) Integrated classification workflow in which positional information (RA, Dec) and spectral-morphological features jointly contribute to celestial object classification.

uncover subtle and previously undetectable patterns in astronomical data has become increasingly urgent. A central aim of this study is to formulate a comprehensive and methodologically robust framework for the spectral classification of celestial objects. Within this framework, the spatial coordinate system of such objects serves as a critical foundation, enabling the detection of distributional trends and physical relationships that might otherwise remain concealed. As illustrated in Fig. 1, the spatial positioning of celestial bodies not only informs their classification but also enhances the interpretability of the resulting astrophysical insights.

One of the most transformative enablers in this domain has been the Sloan Digital Sky Survey (SDSS), a pioneering astronomical initiative that has revolutionized the availability and quality of multi-wavelength observational data (Blanton et al., 2017). Through its extensive and meticulously calibrated datasets, SDSS has provided unprecedented opportunities for the detailed study of both the spectral and morphological properties of a wide range of astrophysical entities. However, as the breadth and complexity of these datasets have expanded, the limitations of conventional classification methods, particularly in terms of efficiency, scalability, and accuracy, have become more pronounced. This has catalyzed the integration of advanced computational paradigms into astrophysical research, most notably the application of machine learning (ML). Among these, astrophysical feature-guided machine learning has emerged as a particularly promising approach, leveraging domain-specific physical principles to enhance the reliability and interpretability of both spectral and Structural classification outcomes (Kotsiantis, 2007; Ian H. Witten, 2016).

The contributions of SDSS extend beyond its extensive spectral catalogs; it has also produced the most detailed three-dimensional mappings of the universe to date, providing the observational foundation for its seventeenth data release (SDSS DR17) (Blanton et al., 2017). Within the scope of this rich dataset, galaxies have been systematically classified into five principal morphological categories: Spiral, Elliptical, Peculiar, Irregular, and Merging, as depicted in Fig. 2. These morphological distinctions are not merely visual taxonomies but are deeply connected to the physical processes that govern galactic formation, evolution, and interaction. Each category encompasses unique structural and dynamical characteristics that, when analyzed in conjunction with spectral features, provide a more comprehensive understanding of the cosmic landscape. The categories are as follows:

- Spiral: Galaxies that exhibit a flat rotating disk containing stars, gas, and dust, with a central bulge composed of older stars. The spiral arms emanate from the center, making them visually distinctive (Hubble, 1936; Kormendy and Kennicutt, 2004).
- Elliptical: These ellipsoidal galaxies have smooth, featureless brightness profiles. They contain older stars and less interstellar matter, resulting in little ongoing star formation (Hubble, 1936; Faber et al., 1996).
- Peculiar: Galaxies that do not fit neatly into other categories, often exhibiting irregular or unusual structures. They may result from interactions or mergers between galaxies (Arp, 1966).
- Irregular: These galaxies lack a regular form like spiral or elliptical ones. They have active star-formation areas and plenty of gas and dust (Hubble, 1936; Hunter and Elmegreen, 2004).
- Merging: These galaxies are in the process of colliding and merging with one another, often resulting in complex and chaotic structures (Barnes and Hernquist, 1992).

The You Only Look Once (YOLO) family, renowned for its real-time object detection capabilities, offers high accuracy and efficiency. YOLOv5, YOLOv7, and YOLOv8, selected for their robustness with large-scale astronomical datasets, were used to classify galaxies into five morphological categories. These classifications demonstrate the variability of galaxy formations and evolution, utilizing high-quality data from multiple celestial regions and advanced technologies such as photometric filters, CCD cameras, and multi-object fiber spectrographs, spanning wavelengths from ultraviolet to near-infrared (Martinazzo et al., 2020; Hasan et al., 2022). Understanding these categories helps explain the creation and development of galaxies and reveals their past, present, and future. Integrating physical information improves the interpretation of the model and provides insight into the underlying astrophysical processes (Ackermann et al., 2018). The key contribution of this study includes:

1. **Exploratory Analysis of Celestial Data:** This research study starts with the SDSS dataset, which offers spectral and morphological data on celestial entities like quasars, stars, and galaxies. Exploratory Data Analysis (EDA) is conducted to identify patterns and enhance the feature set by integrating astrophysical concepts, with a focus on, critical spectral indicators such as redshift and near-infrared filters.

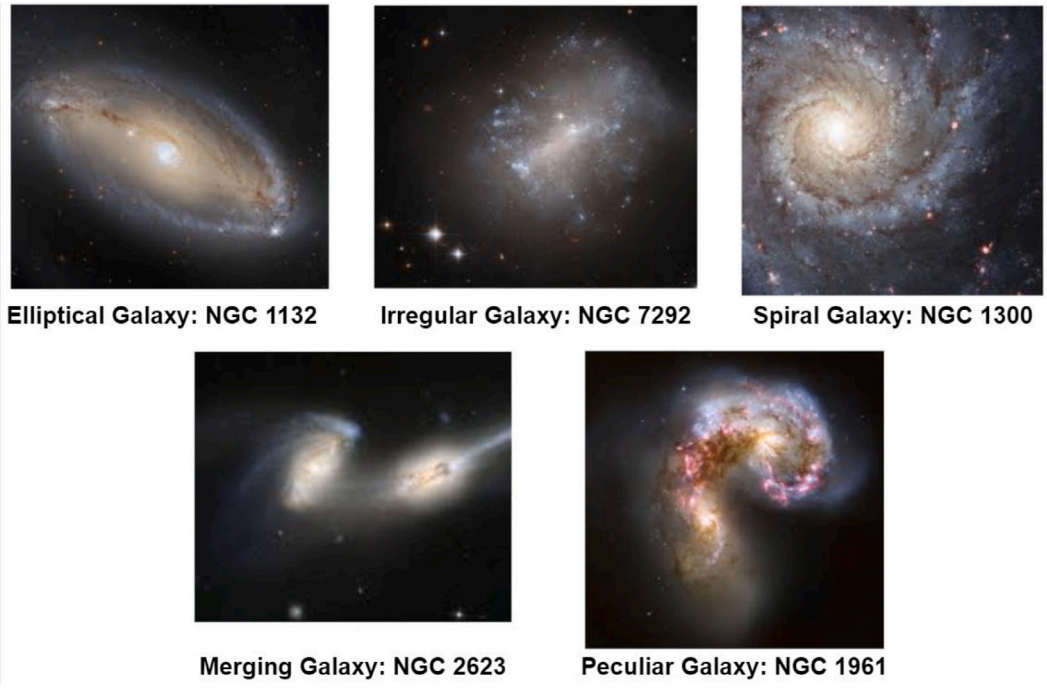


Fig. 2. Morphological galaxy classification. Representative examples of galaxies illustrating key morphological types: elliptical galaxy, irregular galaxy, spiral galaxy, merging galaxy, and peculiar galaxy. All images are courtesy of the NASA/ESA hubble space telescope and the hubble heritage STScI/AURA project (NASA/ESA, 2023).

2. **Preprocessing & Feature Optimization:** The data is subsequently subjected to preprocessing and feature selection, during which critical variables are discovered and extraneous noise is eliminated to facilitate classification. Spectral analysis underpins the classification using profound ML methods such as KNN, SVM, and RF, with the best model identified for categorizing stars, galaxies, and quasars.
3. **Deep Learning (DL) for Galaxy Morphology:** The emphasis subsequently transitions to morphological analysis of the galaxy class that is performed in the previous step. DL models (YOLOv5, YOLOv7, YOLOv8) are trained on selected datasets to classify galaxies into types such as spiral, elliptical, peculiar, irregular, and merging, based on performance evaluation. This framework combines data-driven DL methods with astrophysical concepts to establish a comprehensive system for celestial categorization.

The rest of the paper is structured as follows: The Related Works reviews previous research in the field, focusing on the use of machine learning for classifying astronomical objects. The Methodology outlines the framework for spectral and morphological classification, utilizing SDSS data and models like Random Forest for spectral analysis and YOLO models for morphological classification. Results and Analysis present the performance of these models, with a detailed comparison of their accuracy and effectiveness. Finally, the Conclusion summarizes the findings and suggests future research directions for enhancing classification accuracy with larger datasets.

2. Related works

Numerous studies have effectively utilized machine learning in astronomy for classifying and segmenting objects. Convolutional Neural Networks (CNNs) have been particularly successful in identifying galaxies, stars, and other celestial bodies. Integrating physical knowledge with these approaches has significantly improved their accuracy. However, much remains to be explored in combining ML with feature analysis techniques for astronomical categorization. González et al. (2018) developed a method for automatic detection and classification of galaxies using deep learning and data augmentation. This method used datasets from Galaxy Zoo and the Sloan Digital Sky Survey (SDSS) (Walmsley et al., 2022). Using the YOLO real-time object detection system and the DARKNET framework, the model achieved an accuracy of 80% for the T2 Dataset and 81% for the T4 Dataset. Another study (Pasquet et al., 2019) applied a deep CNN to estimate photometric redshifts and related probability distribution functions (PDF) for galaxies in the SDSS Main Galaxy Sample at z value less than 0.4. The model outperformed earlier ML methods after training on datasets of various sizes.

The application of ML to celestial classification has addressed one of the most persistent challenges in astronomy: sparse and imbalanced datasets (Cody et al., 2024). Traditional methods for celestial classification, such as manual spectral analysis, are labor-intensive and prone to errors when the data volume is limited. However, by identifying patterns and correlations in sparse datasets, ML models have demonstrated impressive effectiveness in getting beyond these restrictions. In Badura et al. (2024), Physics-Informed Neural Networks (PINNs) increase forecast accuracy by including physical restrictions for tasks like orbit determination. Traditional methods use computationally expensive and time-consuming numerical simulations. However, PINNs encode the governing equations of motion as loss functions, ensuring that the model adheres to physical laws during training. PINNs could model star formation, galaxy dynamics, and cosmic ray propagation, revolutionizing computational astronomy beyond orbit determination. The combination of spectral energy distributions (SEDs) with morphological features has also been explored to mitigate degeneracy in the classification of celestial objects. By leveraging both photometric and morphological data, models can achieve more robust classifications, particularly in distinguishing between stars and galaxies (Feng et al., 2024).

In the realm of galaxy classification, several deep learning approaches have shown considerable promise. Jia et al. (2023) introduced a classifier specifically designed for Z-wise and S-wise spiral galaxies. They employed a chirality equivariant residual network (CE-ResNet),

trained on SDSS images with labels from the Galaxy Zoo project. Their model achieved an impressive 92% classification accuracy, far surpassing the typical human accuracy of around 80%. This notable improvement highlights the potential of advanced deep learning models in surpassing traditional methods in certain astrophysical tasks. Similarly, researchers have also explored multimodal networks to classify celestial objects. For instance, a novel network known as MAC-Net combines 2D spectral data and photometric images from LAMOST and SDSS to classify galaxies, quasars, and stars, achieving an outstanding overall accuracy of 98.6% (Zhang et al., 2023). Such models highlight the importance of integrating multiple data types to enhance classification performance.

In another study, Gharat and Dandawate (2022) proposed a deep learning method based on convolutional neural networks (CNN) to categorize galaxies within the SDSS dataset. Their model classified galaxies into ten distinct categories, achieving an accuracy of 84.73% using a transfer learning approach. This result compares favorably with human classification accuracy, demonstrating the potential of CNNs in automating galaxy categorization and potentially reducing the time and effort involved in manual classification. These advancements underscore the increasing importance of machine learning techniques in enhancing the efficiency and precision of astronomical research. YOLO (You Only Look Once) has emerged as a groundbreaking object detection algorithm, gaining widespread adoption across various domains, including astronomy (Redmon et al., 2016). Its ability to predict both bounding boxes and class probabilities in a single pass makes it particularly suited for real-time detection and large-scale image analysis (Imran et al., 2024). This capability is of paramount importance in astronomical research, where datasets can be vast, complex, and constantly evolving (Jocher et al., 2022). The efficiency of YOLO in processing large volumes of astronomical images enables faster insights and improved handling of the high-dimensional data typical in the field.

Galaxy morphology is essential for unraveling the structure, formation, and evolutionary history of galaxies. In early studies, the structural taxonomy of galaxies focused mainly on two broad categories: elliptical and spiral galaxies (Hubble, 1979). This binary system served as the foundation for understanding the diverse shapes and orientations that galaxies could take. However, as observational techniques and data collection methods improved, more categories emerged, leading to the identification of other galaxy types, such as irregular and lenticular galaxies. These additional classifications have significantly deepened our understanding of the diverse range of galaxy structures in the universe. One of the most significant developments in recent years is the ability to identify merging galaxies through advanced image analysis techniques (Hasan et al., 2014). These merging galaxies offer valuable insights into the dynamic processes that occur when two galaxies collide, including the transformation of galactic structures, the triggering of new star formation, and the eventual evolution of the resulting galaxy.

Structural morphology identification does more than just categorize the visual appearance of galaxies; it also helps link these shapes to the underlying physical processes that govern their formation and evolution (Sandage, 2005). By observing and classifying galaxies based on their morphology, astronomers can infer details about the galaxy's age, the conditions of its formation, the role of dark matter, and the interactions that have shaped it over time. The increasing complexity of the classification system, particularly with the addition of sub-classes such as peculiar and merging galaxies, requires more sophisticated methods to discern subtle differences between these categories (Martinazzo et al., 2020). The task of distinguishing these nuanced forms becomes even more challenging as the number of galaxies in observational datasets grows. Fortunately, recent advancements in deep learning techniques have shown remarkable promise in addressing this challenge. These methods, which automatically learn to classify galaxies based on vast amounts of data, enable astronomers to process enormous datasets with greater accuracy and efficiency than ever before. By leveraging deep learning for morphological classification, researchers can now classify galaxies at a scale that was previously unimaginable, making it a transformative tool in modern astrophysics (Martinazzo et al., 2020).

These advancements highlight the effectiveness of combining diverse data modalities with physical principles in machine learning models for astronomical classification tasks. By integrating these elements, the models not only improve in accuracy and efficiency but also offer greater interpretability. This approach establishes a robust framework capable of handling the real-time analysis of large astronomical datasets, thereby pushing the boundaries of current research and enabling deeper insights into the cosmos.

3. Methodology

Astrophysical studies rely on the spectral and morphological characterization of celestial objects to advance our understanding of cosmology. In this work, we utilize data from the Sloan Digital Sky Survey (SDSS) Data Release 17, along with extensive exploratory data analysis (EDA), machine learning, and deep learning techniques, to classify celestial sources based on their spectral and morphological properties. Fig. 3 illustrates the overall workflow of the proposed methodology, which comprises two main components:

Spectral analysis: Stars, galaxies, and quasars are classified using SDSS photometric and spectroscopic measurements in combination with machine-learning algorithms, including k-nearest neighbours (KNN), support vector machines (SVM), and random forests. Following EDA, redshift and infrared features are selected to enhance the discriminative power of the models. The resulting classifiers are rigorously evaluated in terms of accuracy, precision, recall, and F1-score.

Morphological classification: Galaxies are assigned to five morphological categories using a curated set of galaxy images. State-of-the-art object-detection architectures, such as YOLOv5, YOLOv7, and YOLOv8, are trained via transfer learning and fine-tuning to optimize performance. Model quality is assessed using precision, recall, mAP@50, and mAP@50–90, complemented by qualitative inspection of the predicted morphologies. The best-performing model robustly distinguishes spiral, elliptical, peculiar, irregular, and merging galaxies, and the resulting labeled dataset forms a solid basis for subsequent astrophysical investigations.

By integrating spectral classification with advanced deep learning-based morphological analysis, the proposed framework offers a comprehensive and reliable approach for categorizing celestial objects. Its interpretability, accuracy, and scalability make it well-suited for application to larger and more complex astronomical datasets in future studies.

3.1. Exploratory data analysis

After data preparation, exploratory data analysis (EDA) revealed the dataset's fundamental features as shown in Fig. 4. Feature selection methods were employed to identify the most significant characteristics for exact classification. A detailed EDA was performed after data preparation to reveal the dataset's intrinsic properties. Visualizing spectral feature distributions, examining relationships, and detecting outliers are required.

Redshift: Redshift (z) describes the change in wavelength (or frequency) of light due to the relative motion of a light-emitting source and an observer. This phenomenon is most commonly observed in astrophysics to measure the motion of celestial objects. In this section, we derive the redshift in the context of both the Doppler effect and the expansion of the universe.

Redshift is defined as the fractional change in the wavelength of light:

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}}$$

where:

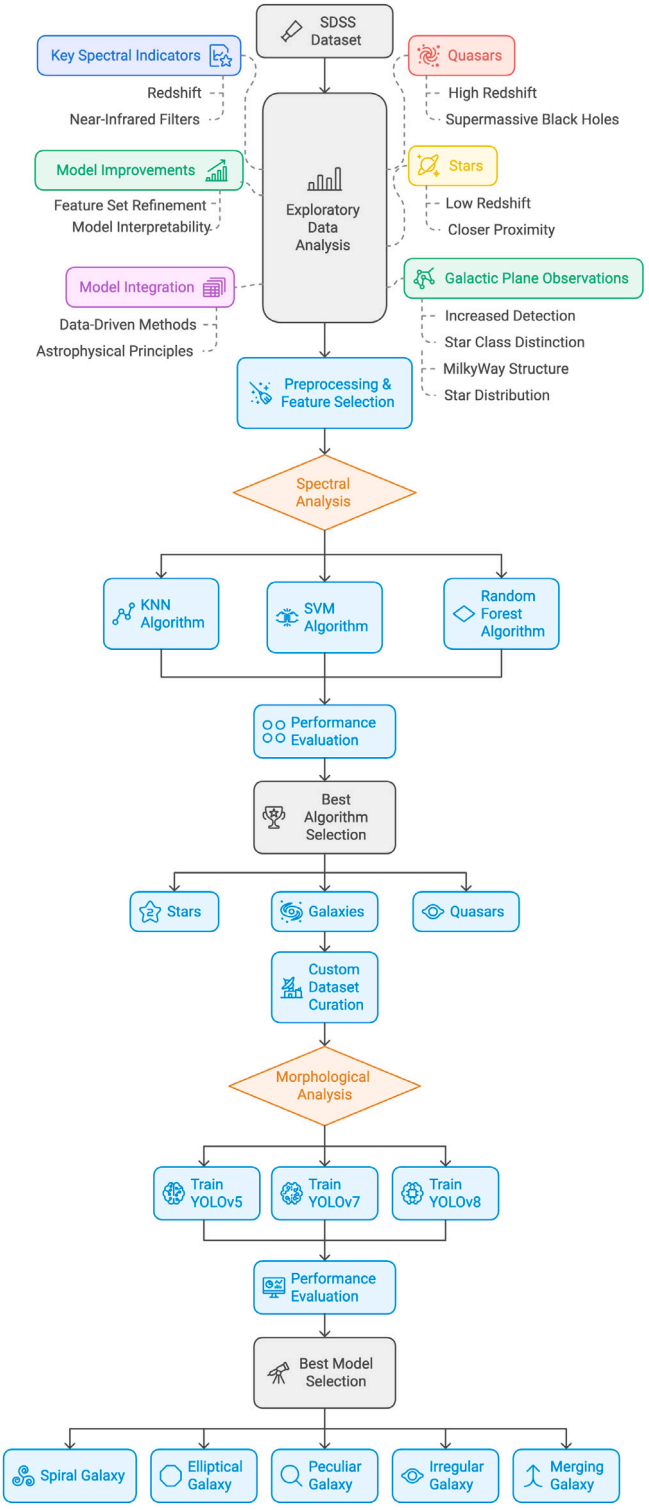


Fig. 3. Overall workflow diagram for the classification.

- λ_{emitted} is the wavelength of light emitted by the source,
- $\lambda_{\text{observed}}$ is the wavelength of light observed after traveling through space.

The relativistic Doppler effect accounts for the relative motion between the source and the observer. The observed wavelength is given

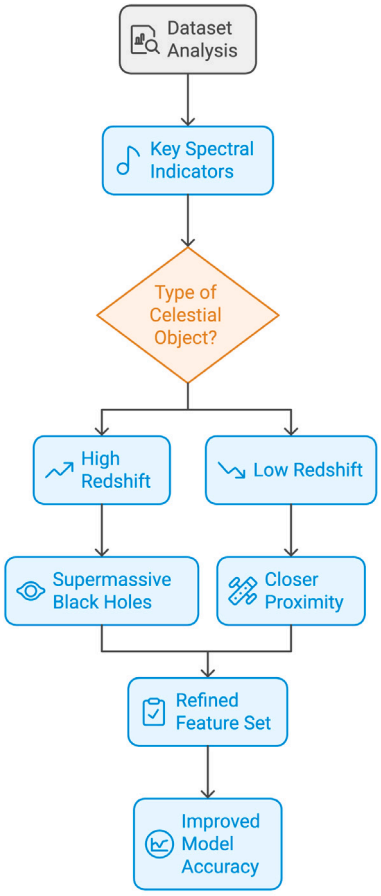


Fig. 4. Data preprocessing and exploratory data analysis.

by Rybicki and Lightman (2024):

$$\lambda_{\text{observed}} = \lambda_{\text{emitted}} \sqrt{\frac{1 + v/c}{1 - v/c}}$$

where:

- v is the relative velocity between the source and observer,
- c is the speed of light in a vacuum.

Substituting this into the definition of redshift:

$$z = \frac{\lambda_{\text{observed}}}{\lambda_{\text{emitted}}} - 1 = \sqrt{\frac{1 + v/c}{1 - v/c}} - 1$$

In an expanding universe, the redshift arises due to the stretching of spacetime (Peebles, 2020). The wavelength of light scales with the scale factor (a) of the universe:

$$\frac{\lambda_{\text{observed}}}{\lambda_{\text{emitted}}} = \frac{a_{\text{observed}}}{a_{\text{emitted}}}$$

where $a_{\text{observed}} = a(t_{\text{observed}})$ and $a_{\text{emitted}} = a(t_{\text{emitted}})$, with t_{observed} and t_{emitted} representing the cosmic times at which the light is observed and was emitted, respectively. The wavelength of photons moving through space stretches proportionally as the universe expands and the scale factor increases with time. Instead of reflecting motion through space, this cosmological redshift represents the expansion of space itself.

Thus, the redshift is given by:

$$z = \frac{a_{\text{observed}}}{a_{\text{emitted}}} - 1$$

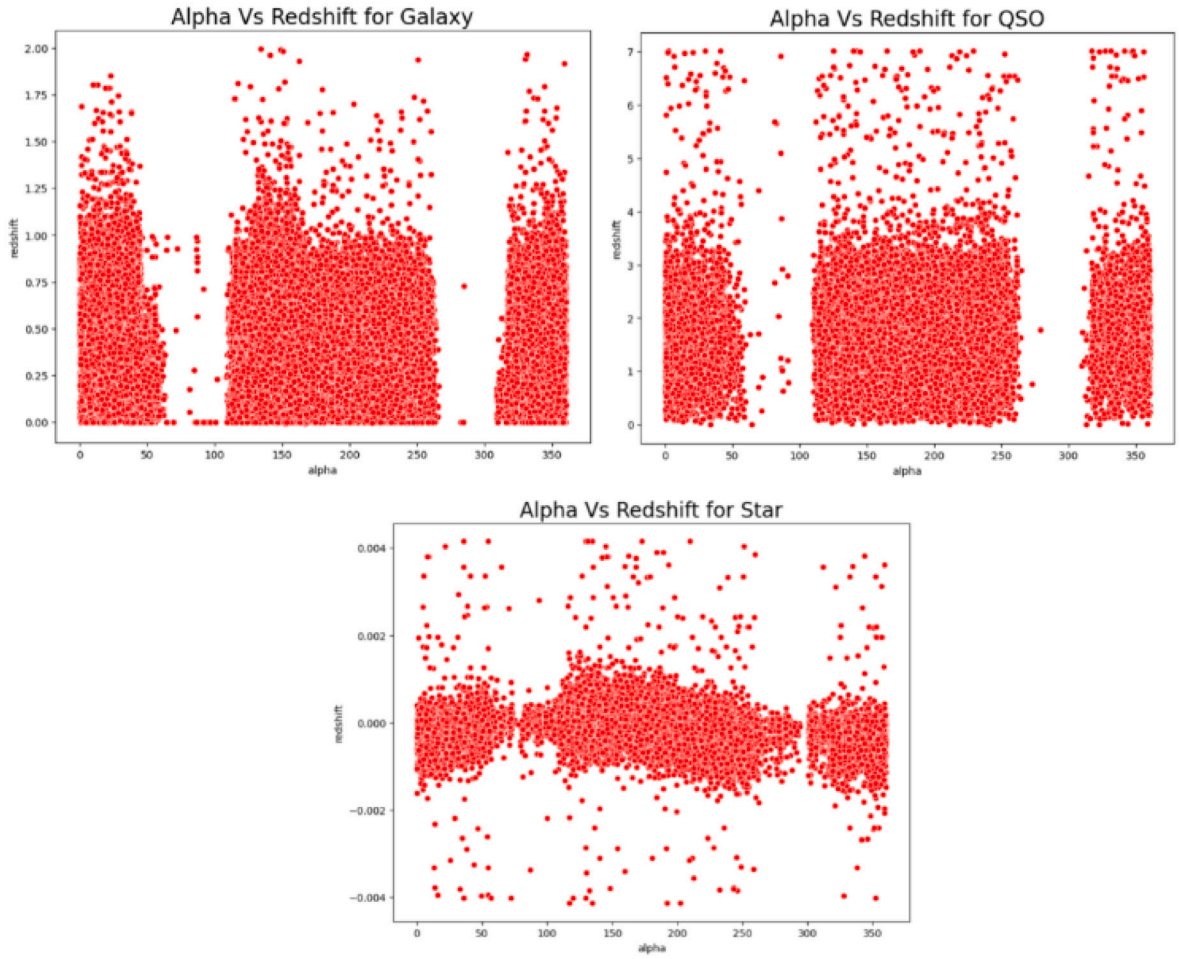


Fig. 5. Redshift for galaxy, quasar, and star classes.

Typically, $a_{\text{observed}} = 1$ (present-day scale factor), so:

$$z = \frac{1}{a_{\text{emitted}}} - 1$$

Redshift analysis will help extract latent spectral information. Through spectral analysis, we scrutinized the distinctive features exhibited by different celestial objects. Fig. 5 shows Redshift for Galaxy, Quasar, and Star Classes, illustrating the distribution of redshift values across the three classes. In our analysis, α (ALPHA) denotes the right ascension (RA) of the object in the celestial coordinate system, i.e. its angular position along the celestial equator, measured in degrees from 0° to 360° (or equivalently in hours from 0 h to 24 h). The plot highlights distinctive patterns in redshift distribution. Notably, we can infer from Fig. 5 that galaxies and quasars manifested prominent redshift values, indicative of their relative interstellar motion and distance, whereas stars exhibited comparatively modest redshift values. This initial observation laid the foundation for subsequent classification efforts.

Spatial Distribution: Analysis of equatorial coordinates revealed distinct spatial distributions for stars, galaxies, and quasars. The scatter plots of Fig. 6 show strong correlations between g and r magnitudes for galaxies and quasars, whereas the star class displays a distinct vertical alignment, indicating less variation in the g-filter compared to the r-filter. Fig. 6 indicates that stars, predominantly located along the galactic plane, exhibited clustering patterns attributed to the Milky Way's structure. In contrast, galaxies and quasars were more evenly distributed, reflecting their extragalactic origins. These spatial patterns offer additional discriminative power, particularly for distinguishing stars from other classes. The findings of the EDA are discussed in the Results and Analysis section.

3.2. Spectral analysis

The spectral analysis of celestial objects is a fundamental task in astrophysics, enabling the categorization and understanding of diverse cosmological phenomena. Using the vast repository of data provided by SDSS17, our overall methodology, Fig. 7, includes meticulous workflows for spectral analysis & classification.

3.2.1. Feature selection

Feature selection techniques were employed to identify the most relevant spectral features that contribute to accurate classification. This step aimed to reduce dimensionality and improve the efficiency and effectiveness of the classification models. The feature selection process revealed compelling correlations between the Class (referring to the type of celestial object) and spectral attributes, particularly redshift and near-infrared spectral values (the i and z filters), as illustrated in Fig. 8. The correlation heatmap displays the Pearson correlation coefficients between all candidate features, where warmer (red) colors denote strong positive correlations and cooler (blue) colors indicate weak or negative correlations. A clear block structure is evident among the photometric bands, with strong correlations between the r, i, and z filters, and a noticeable association between these filters and redshift, consistent with the expected color–redshift trends of galaxies and quasars. In contrast, positional coordinates (alpha, delta) and observation-related identifiers (such as run_ID, cam_col, plate, MJD and fiber_ID) exhibit

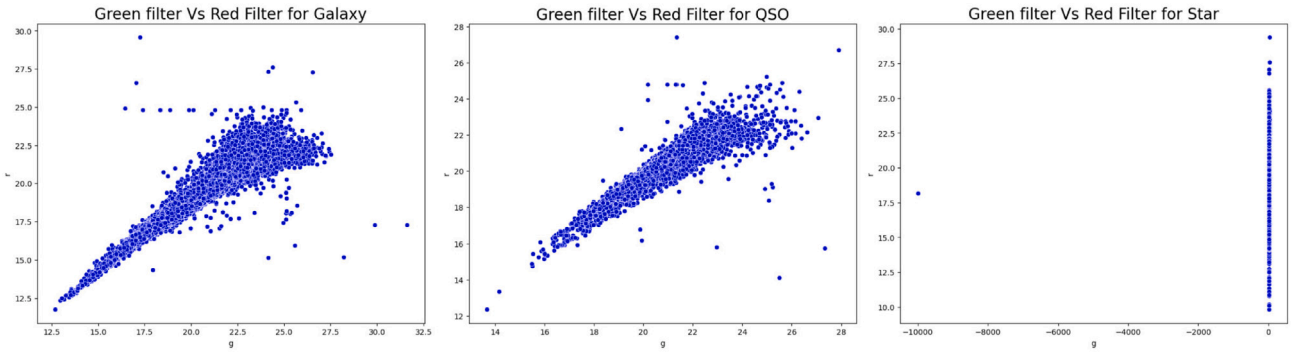


Fig. 6. Amount of light measured by different photometric filters (r vs. g) for galaxy, quasar and star classes.

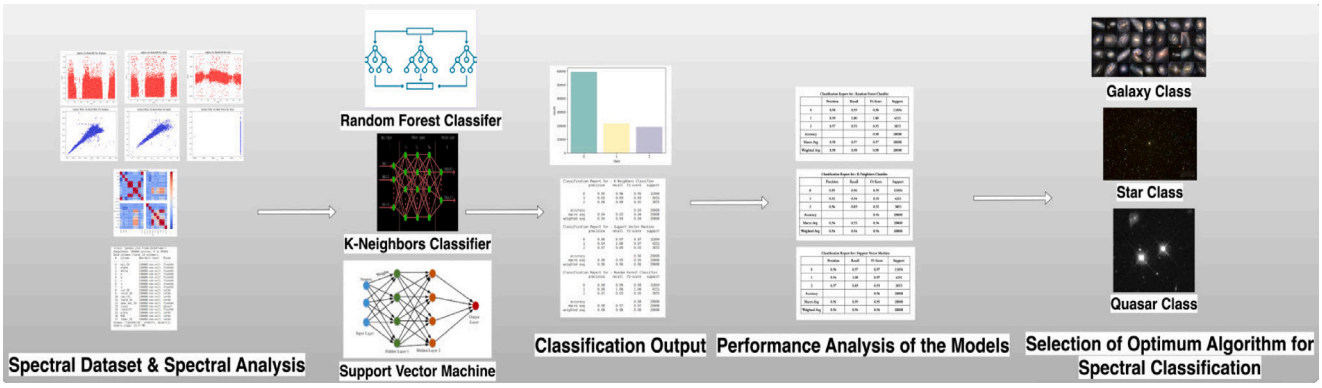


Fig. 7. Workflow diagram for the spectral analysis and classification.

only weak correlations with the class label and with the key photometric features, indicating that they mainly encode acquisition metadata rather than intrinsic astrophysical properties. These patterns support our decision to prioritize redshift and the most informative photometric bands, while de-emphasizing weakly correlated identifier variables, leading to refined feature sets that facilitate robust and physically interpretable classification.

3.2.2. Model training & evaluation

Three frequently used ML models were implemented for spectral analysis and classification: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest Classifier. Each algorithm was chosen for its versatility and effectiveness in classification tasks. Overfitting was avoided by training models on stratified training sets with tuned hyperparameters. The KNN algorithm classifies the data points $\mathbf{x}$ according to the majority class of its nearest neighbors k in the feature space (Cunningham and Delany, 2021). SVM (Cortes, 1995) seeks to maximize the margin between classes by identifying the optimal hyperplane. Random Forest is an ensemble method that mixes numerous decision trees (Breiman, 2001). The mode of all tree forecasts is used to predict the data point $\mathbf{x}$.

To select the best classification model, each model was trained and tested on the preprocessed dataset. After preprocessing, the dataset was divided into training and testing sets using stratified sampling to maintain class proportions. Each ML model was trained with the selected hyperparameters. The performance of each trained model was evaluated on an unseen test set using metrics such as accuracy, precision, recall, and F1-score.

3.3. Morphological classification

A crucial issue in the field of astronomy involves the morphological categorization of celestial objects, such as galaxies that comprehend the structure and evolution of the cosmos (Zhu et al., 2019). In the second phase of the research, the focus shifted to the structural morphology analysis and detection of celestial objects in astronomical images, as shown in the Fig. 9. For this task, we employed the You Only Look Once (YOLO) series of deep learning models (Redmon et al., 2016). Our research aimed to implement and analyze the outcomes of YOLO object detection models using our customized dataset of astronomical object images (Jocher et al., 2022; Wang et al., 2023a; Reis et al., 2023). YOLOv5, YOLOv7 and YOLOv8 models were selected due to their efficiency and fine-tuned on the custom dataset to improve performance on the specific task of galaxy morphology classification (Bochkovskiy et al., 2020). Modern state-of-the-art models often include a relatively large number of layers (Simonyan and Zisserman, 2014), when dealing with challenging classification problems (He et al., 2016).

3.3.1. Dataset preparation

A custom dataset for galaxy morphology classification was curated using images from various astronomical surveys, including SDSS and Galaxy Zoo projects. Images are annotated with bounding boxes to mark regions of interest. The dataset consisted of 1074 images for training and 119 images for testing and validation. The images are classified into five unique categories of galaxies: ‘Elliptical’, ‘Irregular’, ‘Merging’, ‘Peculiar Galaxy’, and ‘Spiral’.

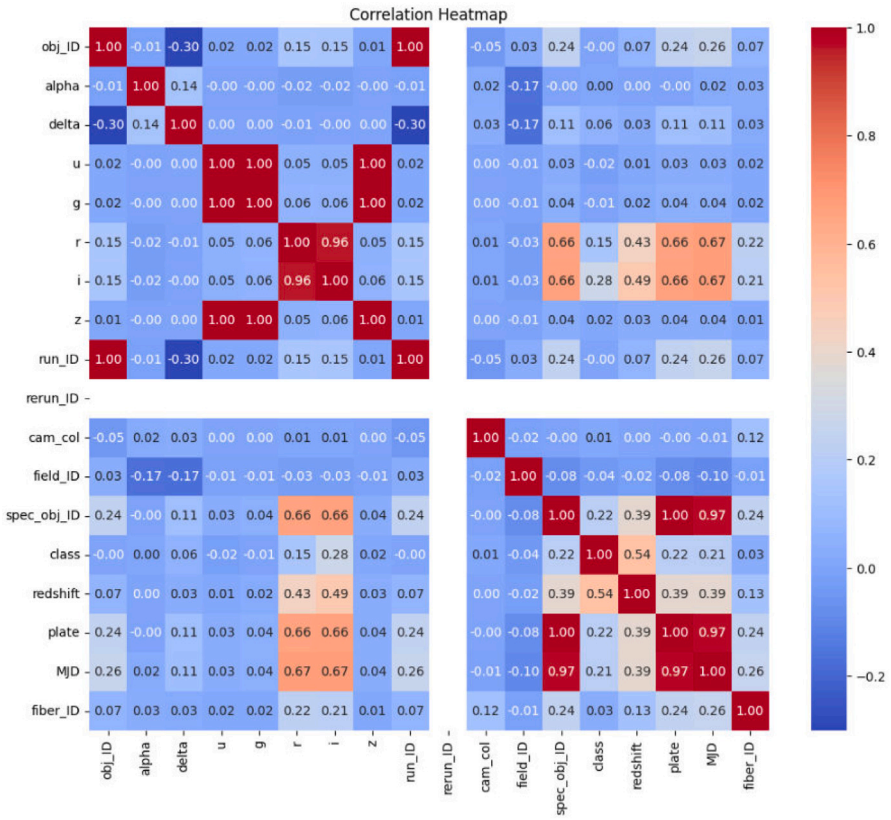


Fig. 8. Correlation heatmap of features.

3.3.2. Model training and evaluation

The YOLO series models were selected for their real-time object detection capabilities, balancing speed and accuracy. YOLOv5, YOLOv7, and YOLOv8 were implemented, each offering improvements in speed and precision. Additionally, hyperparameters such as the learning rate, batch size, and number of epochs were tuned to optimize the model's performance. To initialize these models, pre-trained weights from the COCO dataset (Lin et al., 2014) were used.

To mitigate overfitting on the relatively small morphological dataset, we applied an on-the-fly data augmentation pipeline during training of all YOLO variants. Each mini-batch was generated using random horizontal flips, small rotations, scaling, and translations, which preserve galaxy morphology while varying orientation and apparent size. In addition, moderate brightness, contrast, and color jitter, together with slight Gaussian noise, were used to emulate changes in observing conditions and detector noise, effectively enlarging the training set. Model performance was estimated using stratified cross-validation at the galaxy-class level. After setting aside the independent test set, the remaining images were split into five folds with similar class proportions. For each YOLO variant, four folds were used for training and one for validation in turn; hyperparameters were selected based on the mean validation performance, and the final model was retrained on all training folds and evaluated once on the fixed test set.

Beyond augmentation and cross-validation, several regularization strategies were employed. All YOLO models were fine-tuned from COCO pre-trained weights, which reduces the need to learn low-level features from scratch. Training was performed using stochastic gradient descent with L2 weight decay and early stopping based on validation mAP@50, retaining the best checkpoint. We also adopted lightweight configurations to limit model capacity and further reduce the risk of overfitting. YOLO's loss function combines localization, confidence, and classification losses:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{coord}}\mathcal{L}_{\text{loc}} + \mathcal{L}_{\text{conf}} + \mathcal{L}_{\text{cls}} \tag{1}$$

To accurately detect and classify the complex structures of galaxies, YOLO models for quantitative features were evaluated using a variety of criteria. The Confusion Matrix showed true and false predictions across five morphological categories, showing where the model excelled or struggled. Precision, Recall, and F1-Score were used to assess classification accuracy and dependability, concentrating on the trade-off between true positives and false negatives. The Precision-Recall (PR) curves illustrated the model's behavior across various threshold settings, demonstrating this equilibrium. This evaluation showed that YOLO models can handle vast and complex astronomical information.

4. Results & analysis

The results section comprises three segments. Initially, we will examine the insights from EDA. The findings of the spectrum analysis will be examined along with the influence of EDA on dataset stratification, followed by a discussion of the morphological classification and a comparative study to determine the optimal strategy.

4.1. EDA & spectral classification

EDA helps data-driven researchers to find patterns, correlations, and outliers that influence feature selection. We will examine how EDA affects feature reduction and model optimization, focusing on computational efficiency, hardware requirements, and performance in celestial object classification. As a result of our thorough exploratory

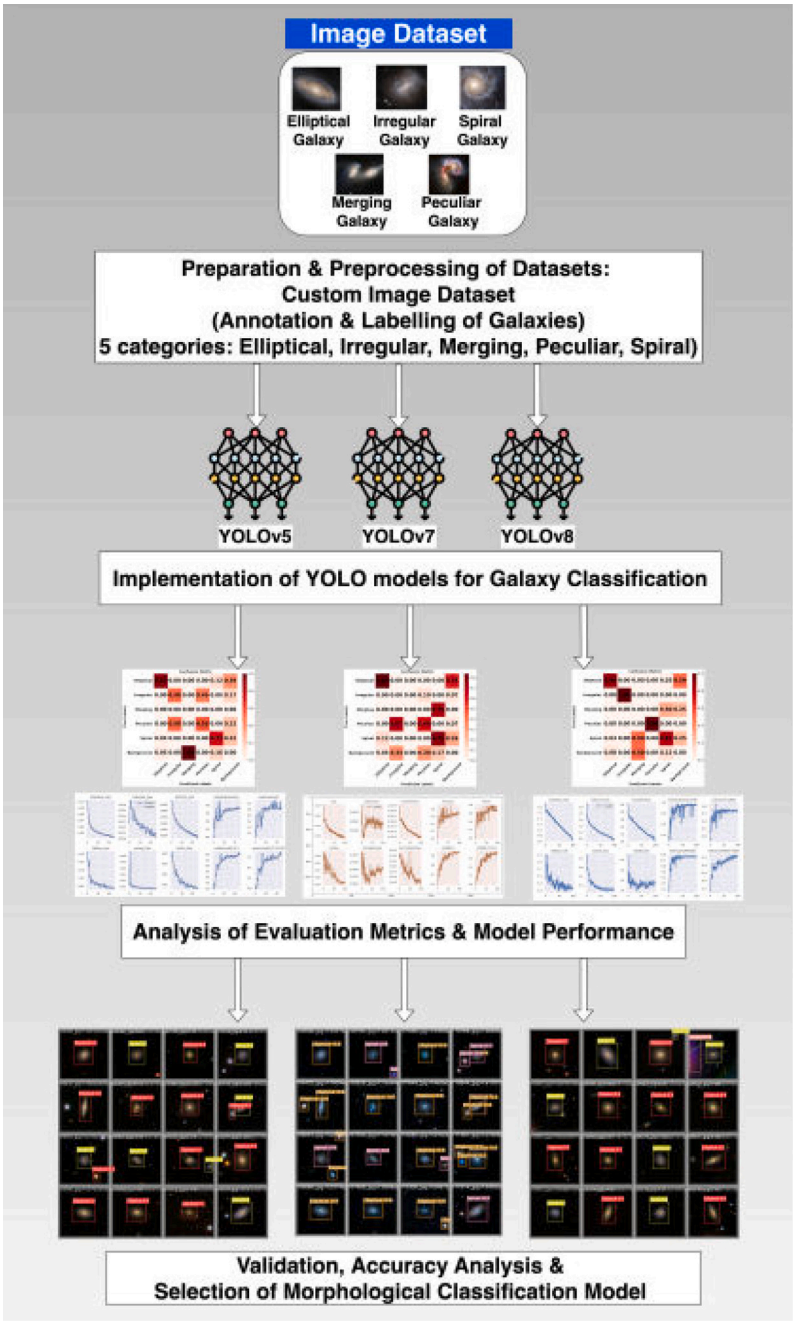


Fig. 9. Workflow diagram for the morphological classification.

evaluation of the dataset shown in Fig. 10, we obtained significant insights pertaining to our dataset:

1. The majority of the objects in the Dataset are Stars and Galaxies, with Quasars making up only 10.58% of samples. Quasars, being relatively scarce, pose a unique challenge due to their distinctive spectral features.
2. Redshift emerged as a critical indicator of celestial object classification. It corresponds to the frequency shift in the electromagnetic spectrum of light emitted by celestial objects, reflecting their motion relative to the observer.
3. Among the classes, quasars exhibited the highest redshift values due to their association with supermassive black holes at the centers of active galactic nuclei, indicating significant interstellar motion and distances. Stars, in contrast, displayed lower redshift values, aligning with their proximity and lower velocities.
4. A unique pattern was observed in the equatorial coordinate distributions of stars compared to galaxies and quasars. Stars exhibited a distribution influenced by the structure of the Milky Way. Observations on the galactic plane yielded a higher probability of detecting stars. This distinction suggests that equatorial coordinates can serve as a discriminative feature in differentiating stars from other celestial classes.

Based on the EDA, the dataset with a reduced feature set is applied for spectral classification. The dataset initially consisted of 17 features, each contributing unique information about celestial objects. Through comprehensive EDA and data analysis, we systematically evaluated the importance and correlation of these features. Redundant, irrelevant, and less informative features were excluded to improve computational efficiency and model performance. As a result, the feature set was reduced to the most relevant attributes: u, g, r, i, z (photometric

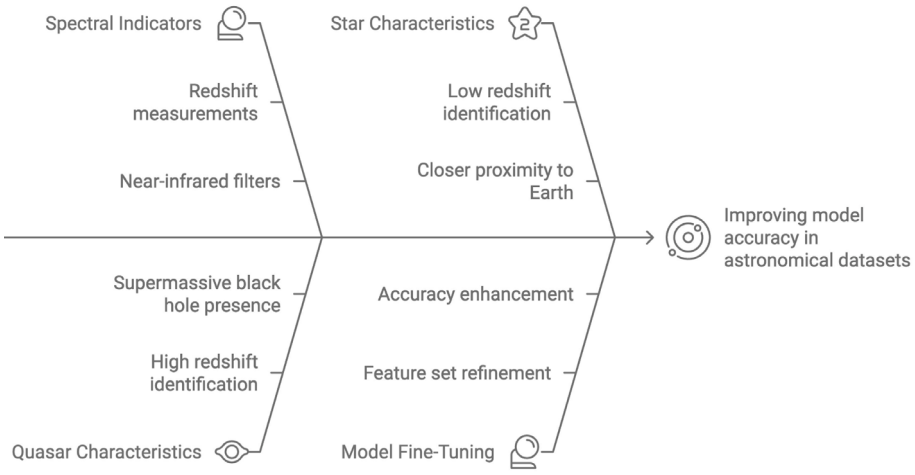


Fig. 10. Key insights from exploratory data analysis.

Table 1
Classification report of three ML classifiers.

ML classifiers	Spectral classes	Precision	Recall	F1-score	Accuracy
K-Neighbors	Galaxy	0.95	0.96	0.95	0.94
	Quasars	0.96	0.89	0.92	
	Star	0.92	0.94	0.93	
SVM	Galaxy	0.96	0.97	0.97	0.96
	Quasars	0.97	0.89	0.93	
	Star	0.94	1.00	0.97	
Random Forest	Galaxy	0.98	0.99	0.98	**0.98**
	Quasars	0.97	0.93	0.95	
	Star	0.99	1.00	1.00	

filters) and redshift, which were found to provide the most significant contributions to classification accuracy.

As discussed in the methodology section, three ML algorithms are applied to classify the reduced dataset. Multiple criteria are used to evaluate the model's accuracy and performance on different spectral bins. Table 1 demonstrates satisfactory accuracy and dependability of all three models, with Random Forest Classifier scoring the highest. Here, Precision, Recall, and F1-Score are reported for each spectral class because these metrics are class-specific and depend on true/false positives and false negatives computed separately for Galaxy, Quasar, and Star. In contrast, accuracy is a global measure calculated over all samples and all classes in the confusion matrix, so it is reported as a single overall value for each classifier. The Random Forest classifier outperforms the other two in accuracy (0.98), precision, recall, and F1-scores across all classes. The Support Vector Machine (SVM) method achieves an accuracy of 0.96. Quasars have lower recall ratings than stars and galaxies, making classification harder for all classifiers.

4.2. Impact of EDA on features reduction and comparative analysis

Key benefits of feature set reduction include lower hardware demand, as it decreases memory and processing requirements, and faster execution through accelerated training and inference processes. It also improves interpretability by simplifying datasets, making it easier to analyze model behavior. Additionally, it enhances performance by removing noise, resulting in more accurate and reliable outcomes. As discussed before, the feature selection process was conducted on 17 attributes, and the final set of selected features was refined to include only the photometric magnitudes in the five bands (u, g, r, i, z) and the redshift, which were deemed most relevant for the classification task.

The reduced feature set performs comparably or slightly better than the full feature set in terms of precision and recall when using Random Forest and SVM classifiers as shown in Table 2. It also significantly improves computational efficiency by reducing memory usage and speeding up execution times. Among classifiers, Random Forest consistently achieves the best performance, with an overall accuracy of 98%, outperforming other models across all classes and feature sets. SVM achieves a reliable 96% accuracy, although it struggles with Quasar recall. K-Neighbors performs the worst, particularly with the full feature set. Class-specific observations reveal that Quasars have lower recall across all classifiers, suggesting a need for improved feature representation or additional preprocessing. In contrast, Stars and Galaxies are classified with high precision and recall scores across all models.

Feature selection greatly improved model performance and accuracy. When employing a smaller feature set, K-Neighbors, which is sensitive to irrelevant information, improved precision, recall, and F1 Scores, especially for the Star and Quasar classes. Random Forest and SVM models performed well in both instances, demonstrating their robustness to irrelevant features. Streamlining computations through feature reduction resulted in a 46.8% decrease in memory usage and a 45.2% reduction in execution time, making the process faster and more suitable for resource-constrained environments (see Fig. 11). This decrease in feature sets shortened experimentation cycles, allowed the study of larger datasets or the integration of additional ML models, and made the approach possible for hardware-limited systems, such as edge devices.

A separate individual comparative analysis is presented in Fig. 12. Using the reduced feature set, the K-Nearest Neighbors classifier improves accuracy and recall for the Galaxy class, with precision increasing by 6.0% and recall remaining unchanged. Galaxy classification is effective with the Random Forest classifier across both feature sets. Quasars exhibit a significant improvement in recall for the K-Nearest Neighbors classifier, with an increase of 7.0%. The performance measures of SVM and Random Forest classifiers do not change significantly in this category. With a smaller feature set, the K-Nearest Neighbors classifier improves recall in the Star class from 81% to 94%. The Random Forest classifier excels in star classification, achieving an F1-score of 100%, which proves its proficiency.

The decreased feature set demonstrates significant enhancements in classification performance across multiple criteria. For galaxies, precision increased by 6.7%, although recall remained constant. Quasars

Table 2
Quantitative and comparative analysis of the classifiers.

Classifier	Feature set	Classes	Precision	Recall	F1-score	Accuracy (%)	Improvement
K-Neighbors	Full set	Galaxy	0.89	0.96	0.92	**89.3**	**4.9%**
		Quasar	0.96	0.82	0.89		
		Star	0.90	0.81	0.85		
	Reduced set	Galaxy	0.95	0.96	0.95	**93.7**	
		Quasar	0.96	0.89	0.92		
		Star	0.92	0.94	0.93		
SVM	Full set	Galaxy	0.96	0.97	0.97	**94.6**	**0.7%**
		Quasar	0.97	0.89	0.93		
		Star	0.95	0.98	0.97		
	Reduced set	Galaxy	0.96	0.97	0.97	**95.3**	
		Quasar	0.97	0.89	0.93		
		Star	0.94	1.00	0.97		
Random Forest	Full set	Galaxy	0.98	0.99	0.98	**97.8**	**0.2%**
		Quasar	0.97	0.93	0.95		
		Star	0.99	1.00	1.00		
	Reduced set	Galaxy	0.98	0.99	0.98	**98**	
		Quasar	0.97	0.93	0.95		
		Star	0.99	1.00	1.00		

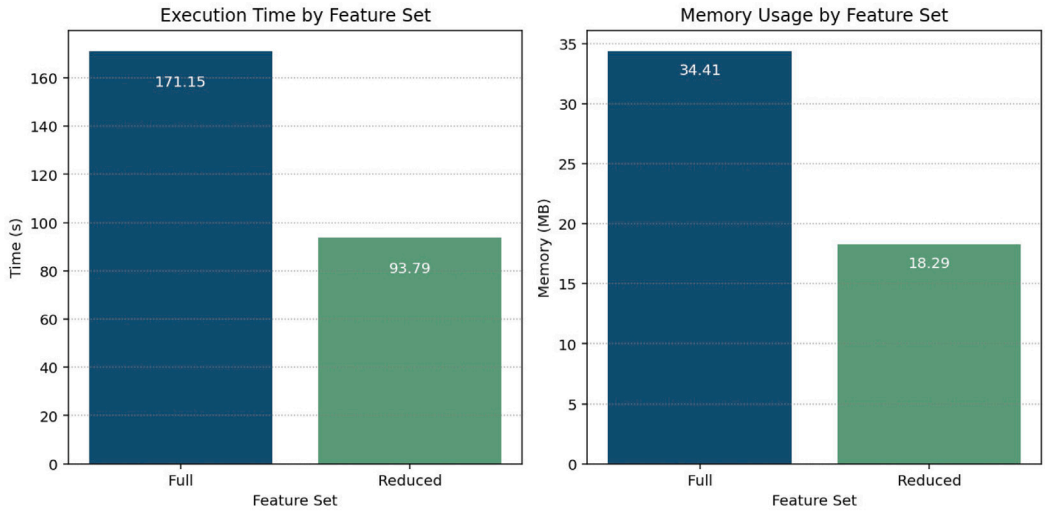


Fig. 11. Execution time and resource usage comparison between full and reduced feature sets.

experienced a precision enhancement of 1.0% and a significant memory augmentation of 8.5%. For stars, both precision and recall exhibited significant improvements, with precision rising by 2.2% and recall by an astonishing 23.5%. The diminished feature set results in a 4.4% enhancement in accuracy, underscoring its efficacy in providing precise outcomes with less processing requirements.

Significant enhancements in classification performance and computational efficiency were demonstrated by EDA-driven feature selection procedures. The systematic reduction to six key criteria elucidated the significance of informed dimensionality reduction for resource optimization and scalable model development, hence underscoring the transformative impact of EDA in astrophysical research.

4.3. Results of morphological analysis

We analyze the performance measures and loss functions of YOLOv5, v7, and v8 models in this section. The algorithms were trained on a proprietary dataset for morphological categorization of spiral, elliptical, peculiar, irregular, and merging galaxies. The evaluation considers precision, recall, mAP, and loss component progression during training. These measurements demonstrate the models' ability to detect and classify celestial objects based on their morphology.

As shown in Fig. 13, all three models show a steady decrease in both training and validation losses, indicating effective learning. YOLOv8, however, shows the most rapid decrease, suggesting faster and more efficient learning. YOLOv8 outperforms both YOLOv5 and YOLOv7 in terms of precision and recall. While YOLOv5 shows good improvement, YOLOv7 is slightly better, but YOLOv8 stands out with the highest precision and recall metrics, making it the most reliable model for object detection with fewer false positives and more true positives. The mAP@0.5 score increases progressively from YOLOv5 to YOLOv8, with YOLOv8 achieving the highest score. This indicates that YOLOv8 has the best object detection capability at the lower IoU threshold. YOLOv8 also excels in this metric (mAP@0.5:0.95), suggesting that it has the best overall performance in detecting objects across a range of IoU thresholds. This makes it the most robust model among the three for general object detection tasks.

Confusion matrices of YOLOv5, YOLOv7 and YOLOv8 models are also shown in Fig. 14 for further analysis. The figure reports a normalized confusion matrix (values in [0,1]), where diagonal cells indicate

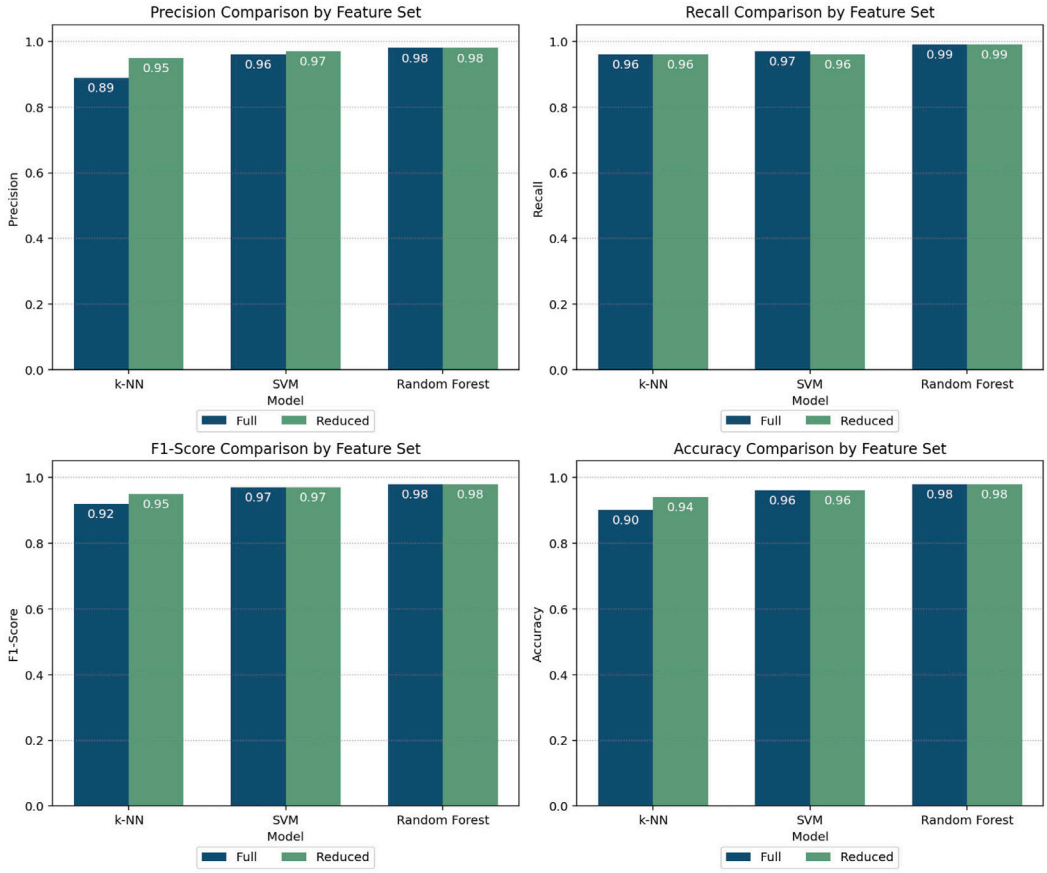


Fig. 12. Performance metrics comparison across models for full and reduced feature sets.

the per-class correct classification rate. For the Elliptical class, the diagonal value is 0.92, i.e., YOLOv5 correctly identifies 92% of Elliptical galaxies but struggles with rarer classes like peculiar and merging galaxies, often misclassifying them due to limited feature representation and dataset biases. YOLOv7 improves detection for spiral and elliptical galaxies but misclassifies 67% of peculiar galaxies as elliptical and frequently confuses merging galaxies with spiral galaxies. YOLOv8 performs best, achieving 89% accuracy for spirals and near-perfect results for elliptical and irregular galaxies. Misclassifications are minimal, with only 3% of spirals misidentified as merging; however, some confusion persists regarding the peculiar and merging classes. YOLOv8 emerges as the most robust and reliable model for complex detection tasks, while YOLOv7 and YOLOv5 offer more resource-efficient alternatives with some trade-offs in performance.

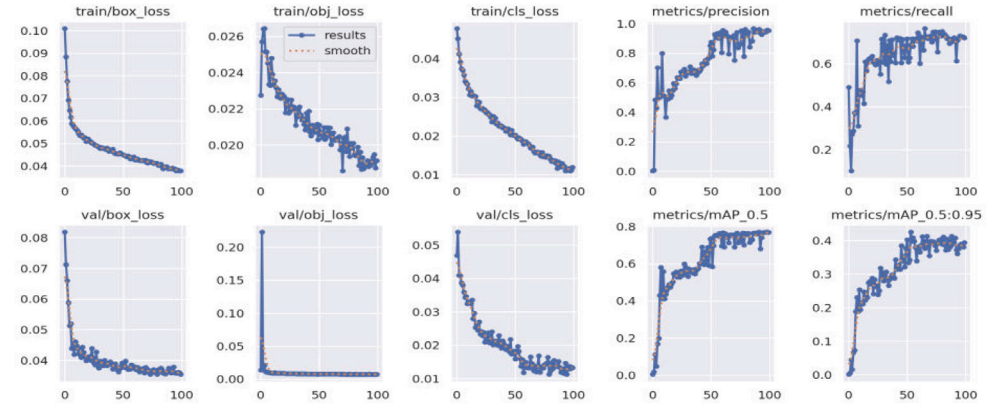
Table 3 shows that YOLOv5 performed well in the Irregular and Peculiar Galaxy classes, with near-perfect precision and recall. Precision for all classes was 92.6% and recall was 68.9%. YOLOv7 had poorer precision (68.7%) and recall (66.4%), but enhanced recognition of Peculiar and Spiral galaxies, with a better balance in dense or complex images. The Irregular Galaxy class severely limited YOLOv7's precision. Among the three models, YOLOv8 performed best, with 95.5% precision and 73.7% recall across all classes. With high mAP50 and mAP50-95 scores, it classified Elliptical, Irregular, Peculiar, and Spiral galaxies well. The model outperformed YOLOv5 and YOLOv7 in detecting uncommon or complicated galaxy formations and common morphological categories.

The Merging Galaxy class challenged models. Our model could not learn from the training data because merging galaxies are rare. Another study (Wang et al., 2023b) utilized a different dataset and data augmentation techniques (e.g., affine transformation, filtering) to classify galaxy morphology, identifying spiral, elliptical, lensed, and stellar objects. After 100 epochs, YOLOv8 achieved a peak mAP@0.5 of 0.767 in the Peculiar Galaxy and Irregular classes, outperforming in both precision and recall. YOLOv7 followed closely with mAP@0.5 of 0.639, excelling in Spiral and Peculiar Galaxy but faltering in Merging Class. The lowest mAP@0.5 was 0.539 for YOLOv5, which performed moderately across classes but struggled to detect Merging and Irregular forms. YOLOv8 was most balanced, especially for complex galaxy classifications.

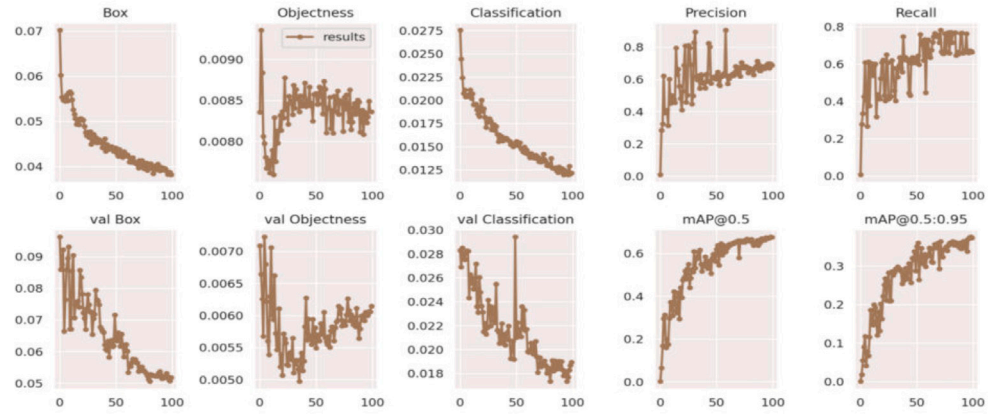
Precision–recall and F1-confidence curves for different galaxy types are shown in Fig. 15. The graphs show precision–recall trade-offs at class-specific levels. AUC and mAP metrics reveal classification performance. Irregular galaxies have a near-perfect score (mAP 0.995), indicating high precision and recall. Peculiar galaxies also perform well (mAP 0.995). Merging galaxies perform poorly (mAP 0.000), possibly due to low confidence forecasts or data discrepancies. The blue curve indicates modest mAP (0.767) across all galaxy types, reflecting the general average. These graphs show the precision–recall balance of the F1-score as a function of confidence levels. Different galaxy types exhibit distinct peaks, indicating their confidence level in maximizing F1-score.

4.4. Morphological detection results and analysis

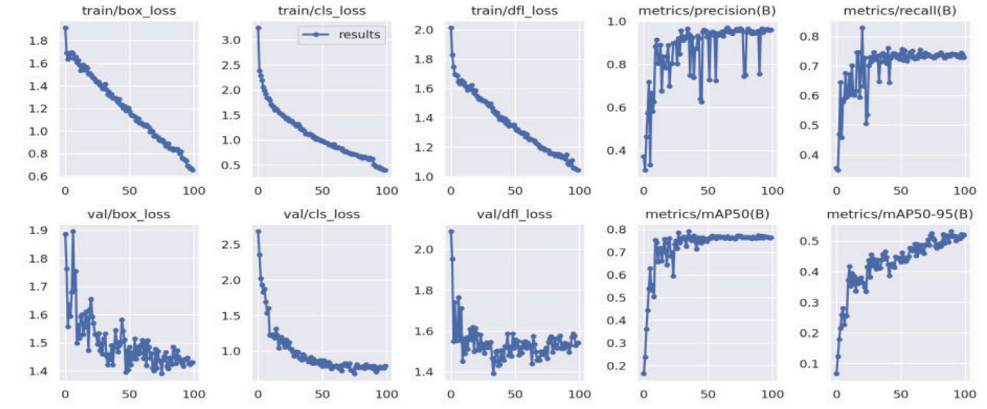
Table 4 shows the final detection results combined from all 5 detected classes. YOLOv5, Fig. 16a, showed the most variability in its results, with a higher rate of false positives and lower mAP scores, particularly in the mAP50-90 range. This suggests that YOLOv5 might



(a) YOLOv5

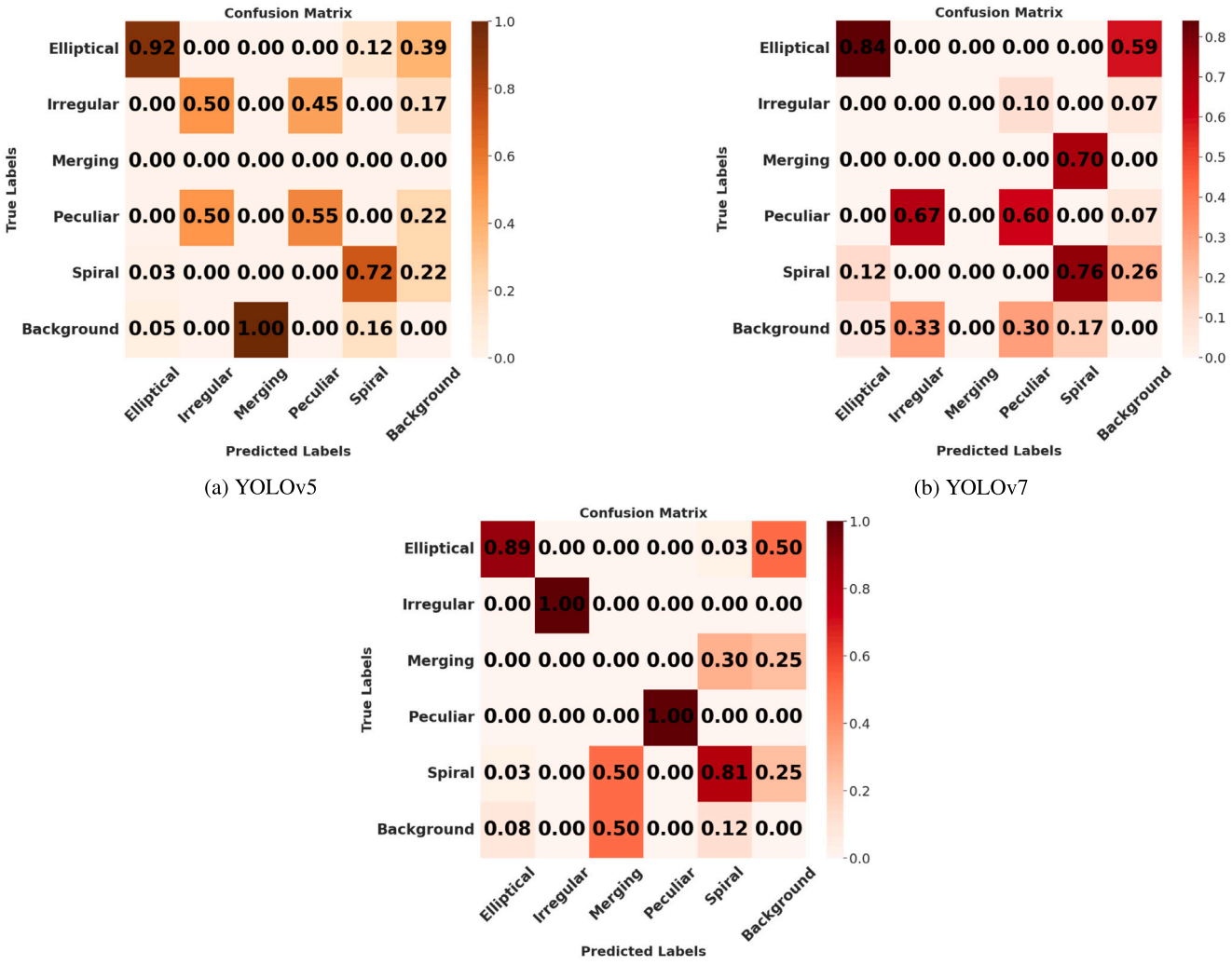


(b) YOLOv7



(c) YOLOv8

Fig. 13. Model performance metrics and loss curves of the YOLO models. The horizontal axis in every small panel is the training epoch $(0-100)$. The vertical axis is the value of the metric in each Top row: box(box_loss), objectness(obj_loss), classification(cls_loss) and Bottom row: val box(box_loss), val objectness(obj_loss), and val classification(cls_loss) show the training and validation losses for bounding-box regression, objectness score, and class prediction (smaller is better). Precision and recall show the corresponding training precision and recall values between 0 and 1 (higher is better). mAP@0.5 and mAP@$0.5:0.95$ are the validation mean average precision at IoU 0.5 and averaged over IoU $0.5-0.95$, also between 0 and 1 (higher is better).



(a) YOLOv5

(b) YOLOv7

(c) YOLOv8

Fig. 14. Confusion matrices for YOLOv5, YOLOv7, and YOLOv8 models.

Table 3
Performance evaluation of YOLO V5, V7 and V8 models (100 Epochs) for five galaxy classes individually and combined.

Evaluation	Precision			Recall			mAP50			mAP50-95		
Classes\YOLO models	v5	v7	v8	v5	v7	v8	v5	v7	v8	v5	v7	v8
Elliptical galaxy	0.777	0.694	0.933	0.895	0.947	0.842	0.908	0.939	0.928	0.341	0.410	0.419
Irregular galaxy	1.000	0.095	0.94	0.956	0.5	1.000	0.995	0.566	0.995	0.721	0.460	0.846
Merging galaxy	1.000	1.000	1.000	0	0	0	0.029	0.021	0	0.009	0.014	0
Peculiar galaxy	1.000	0.916	0.972	0.872	1.000	1.000	0.995	0.980	0.995	0.640	0.575	0.842
Spiral galaxy	0.852	0.730	0.931	0.721	0.875	0.842	0.830	0.878	0.947	0.416	0.404	0.551
All classes	0.926	0.687	0.955	0.689	0.664	0.737	0.751	0.677	0.773	0.425	0.372	0.532

struggle with the nuances of galaxy classification, possibly due to its older architecture and less sophisticated feature extraction capabilities. YOLOv7, Fig. 16(b), performed well, but had a slight drop in recall, indicating that, while it was precise, it occasionally missed detecting certain galaxies, especially those with less distinct morphological features. This balance between precision and recall made YOLOv7 a reliable but slightly less confident model compared to YOLOv8.

YOLOv8, Fig. 16(c), consistently outperformed the other versions, particularly in precision and mAP scores, indicating its superior ability to correctly identify and localize the galaxy classes with minimal false positives. The improvement in YOLOv8 can be attributed to its enhanced architecture, which enables better feature extraction and more precise localization.

YOLOv8 outperforms YOLOv7 and YOLOv5 in accuracy and confidence, particularly for Spiral and Elliptical galaxies, maintaining consistent confidence levels (0.7–0.9) even in complex or overlapping scenarios. It effectively balances precision and recall, reducing false positives and negatives. YOLOv7 achieves good accuracy but shows greater variability in confidence (0.3–0.7), especially in crowded scenes. While reliable for single-object detection, it struggles with overlapping or densely packed objects. YOLOv5 is the least consistent, with confidence

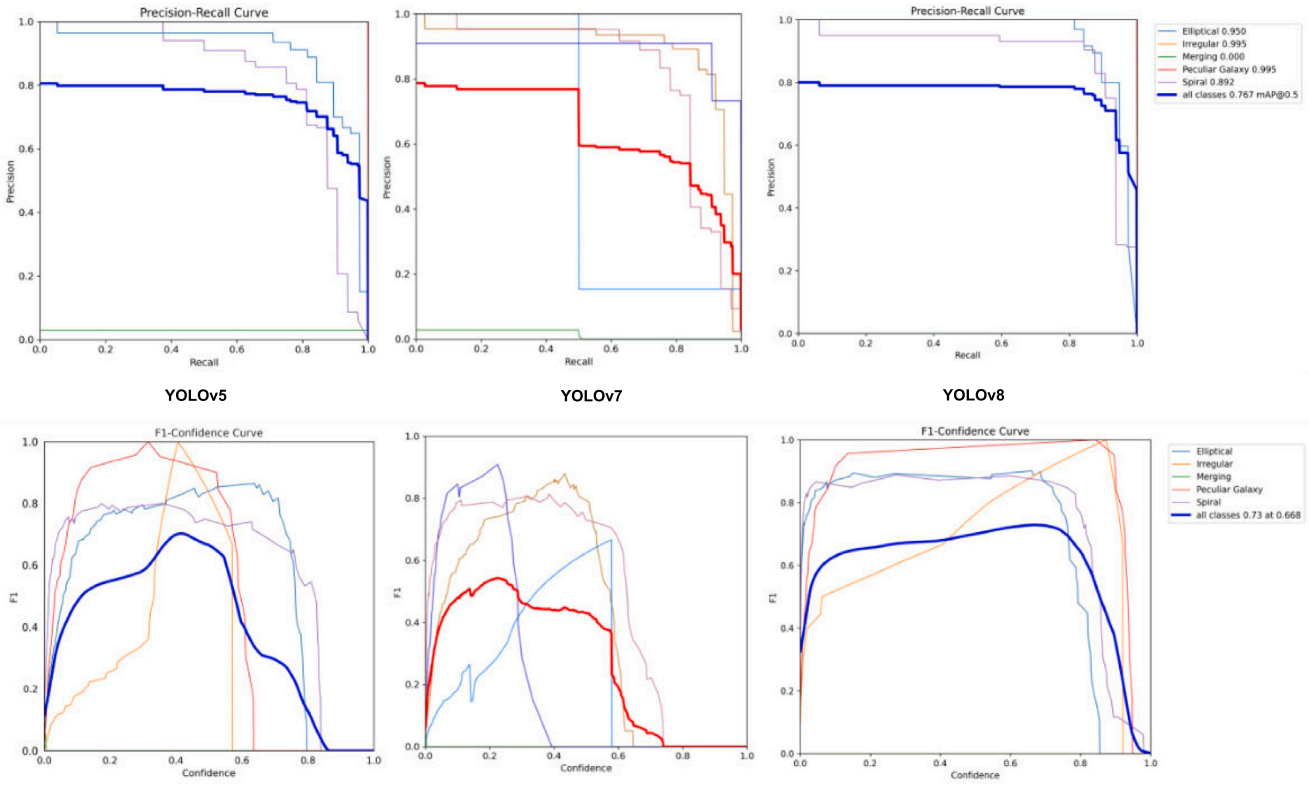


Fig. 15. Precision-recall curves of YOLOv5, YOLOv7, YOLOv8 (Top) and F1-confidence curves of YOLOv5, YOLOv7 and YOLOv8 (Bottom).

Table 4
Final detection metrics of YOLO models (v5, v7, v8).

Metric	YOLOv5	YOLOv7	YOLOv8
Box loss	**0.80**	**0.75**	**0.60**
Objectness loss	**0.25**	**0.20**	**0.15**
Classification loss	**0.15**	**0.18**	**0.12**
Precision	**0.70**	**0.72**	**0.78**
Recall	**0.68**	**0.70**	**0.75**
mAP50	**0.80**	**0.80**	**0.90**
mAP50-95	**0.30**	**0.35**	**0.50**

scores ranging from 0.4 to 0.8. Its performance drops notably in complex settings, reflecting limitations in handling dense or overlapping galaxies. YOLOv8 is the most robust and reliable model for morphological classification, excelling in distinguishing Spiral and Elliptical galaxies and handling complex images. YOLOv7 is a viable alternative but less effective in challenging scenarios, while YOLOv5 is suited for simpler tasks with fewer morphological complexities.

5. Conclusion

This study addresses the challenging problem of celestial object classification using astrophysically guided features and advanced neural networks. Traditional machine learning techniques like k-NN, SVM, and Random Forest were effective in classifying stars, galaxies, and quasars using SDSS spectral data, with Random Forest achieving the highest accuracy. YOLOv5, YOLOv7, and YOLOv8 were employed for galaxy morphological classification, with YOLOv8 emerging as the most effective, delivering high precision in detecting subtle morphological features. However, the Merging Galaxy class, due to its rarity, posed challenges for training, as the model struggled with the limited data. Data augmentation was applied in this study; however, a more rigorous augmentation process will be implemented in future work, along with data injection and transfer learning, to address the challenges posed by small datasets for rare classes. The results demonstrate the transformative potential of machine learning in the real-time processing of large astronomical datasets, laying the groundwork for future automated categorization pipelines that integrate physics and machine learning to solve complex astrophysical challenges and deepen our understanding of the universe.

CRediT authorship contribution statement

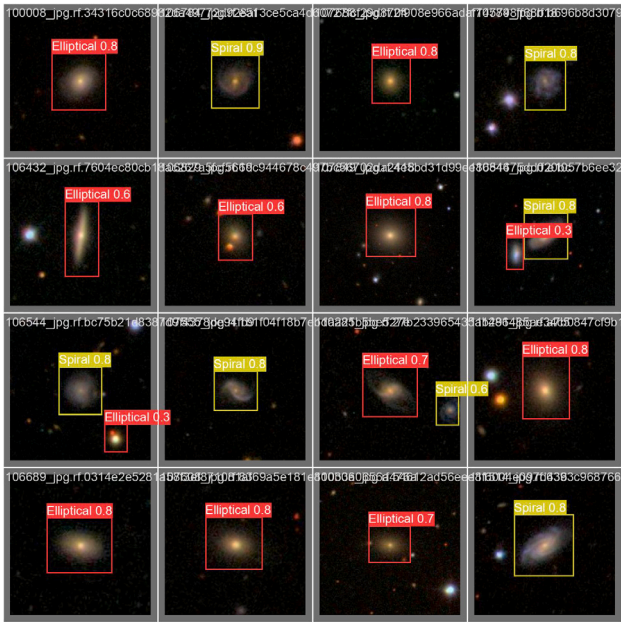
Md. Fairuz Siddiquee: Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Md Mehedi Hasan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Conceptualization. **Shifat E. Arman:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis. **Md. Shahedul Islam:** Project administration, Methodology, Investigation, Conceptualization. **AKM Azad:** Validation, Resources, Methodology, Investigation.

Declaration of competing interest

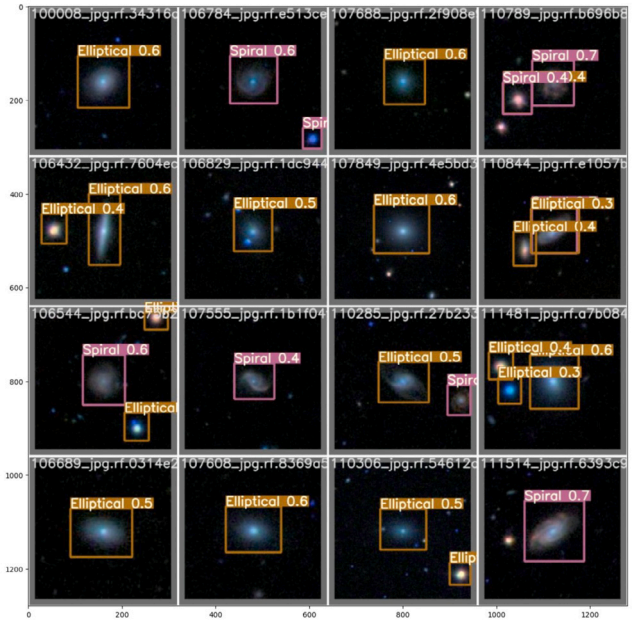
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

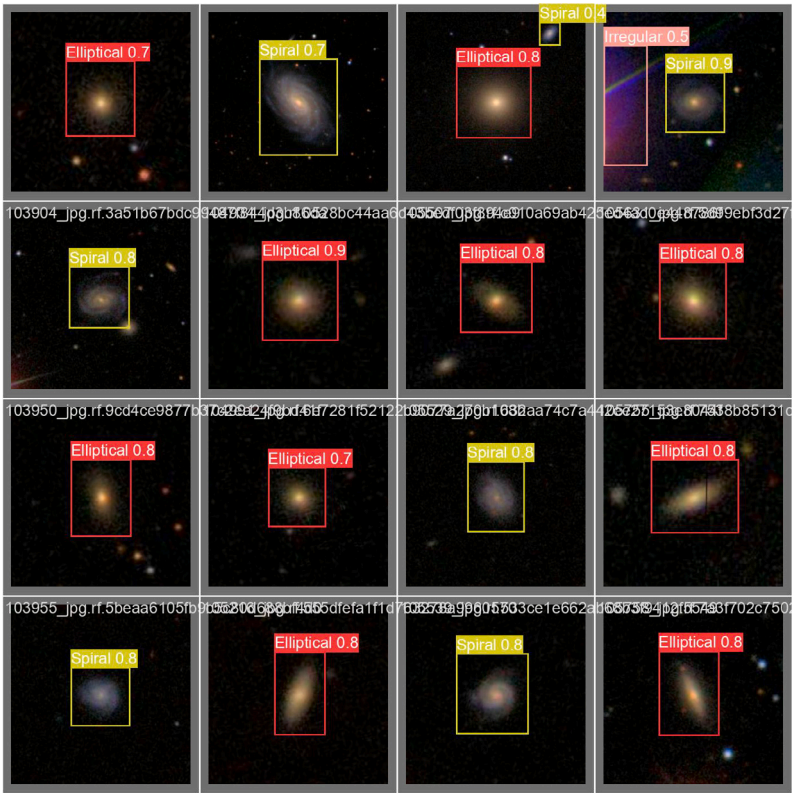
Data will be made available on request.



(a) Detection results using YOLOv5.



(b) Detection results using YOLOv7.



(c) Detection results using YOLOv8.

Fig. 16. Comparison of galaxy detection outputs produced by (a) YOLOv5, (b) YOLOv7, and (c) YOLOv8 models. The visualization highlights improved bounding-box precision and class differentiation in newer versions.

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